



Spring Framework

Training Assignment

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RECORD OF CHANGES

No	Effective Date	Change Description	Reason	Reviewer	Approver
1	26/Sep/2024	Create a new assignment	Create new		

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		CODE	:	JSFW_M_A102
		TYPE	:	MEDIUM
		LOC	:	N/A
		DURATION	:	120 minutes

Unit 1: Spring Framework Core

Objectives:

After completing this assignment, you will:

1. Understand and practice Object-Oriented Programming (OOP) and Input/Output (IO) operations in Java.
2. Understand and apply the core concepts of the Spring Framework, such as Inversion of Control (IoC) and Dependency Injection (DI).

Problem Descriptions:

We are beginning to build our Learning Management System (LMS). This task involves creating the core Course entity for our "course-sharing" system and a service to save/load course data from a file, all managed by the Spring container.

Task 1: Course Class

Create a class named **Course.java** in the model package that:

- Implements the Serializable interface.
- Contains the following attributes:
 1. String title
 2. String instructorName
 3. String description
 4. int durationHours
- Includes a constructor with the above attributes as parameters.
- Implements a method void **displayInfo()** that outputs the course details. For example: [Course: Spring Framework] by John Doe (40 hours): Learn Spring Core, MVC, and Boot.

Task 2: CourseService Class

Create a class named **CourseService.java** in the service package that contains:

- A method void **saveCourses**(List<Course> courses) that saves a list of courses to a file named courses.dat. (Hint: Use **ObjectOutputStream**).
- A method List<Course> **getCourses**() that reads the contents of courses.dat and returns the list of courses. (Hint: Use **ObjectInputStream**).

Task 3: Test Class

Create a class named **Test.java** in the main package that uses the **CourseService** to perform the following:

- An attribute `CourseService` `courseService` for the IO operations.
- In the **main()** method:
 1. Create a list of `Course` objects and save the list to the file `courses.dat` using the **saveCourses()** method from `courseService`.
 2. Retrieve the saved list of courses from `courses.dat` using the **getCourses()** method and call the **displayInfo()** method for each course in the list.

Notice: You must configure Spring to create a bean for the **CourseService** class and inject that bean into the `courseService` attribute of the `Test.java` class (using either XML configuration or Annotation-based configuration).

Technical Requirements:

- JDK 1.8 +
- Maven 3 +
- Spring Framework Core

-- THE END --